

# OOPS PROJECT

# PRESENTATION

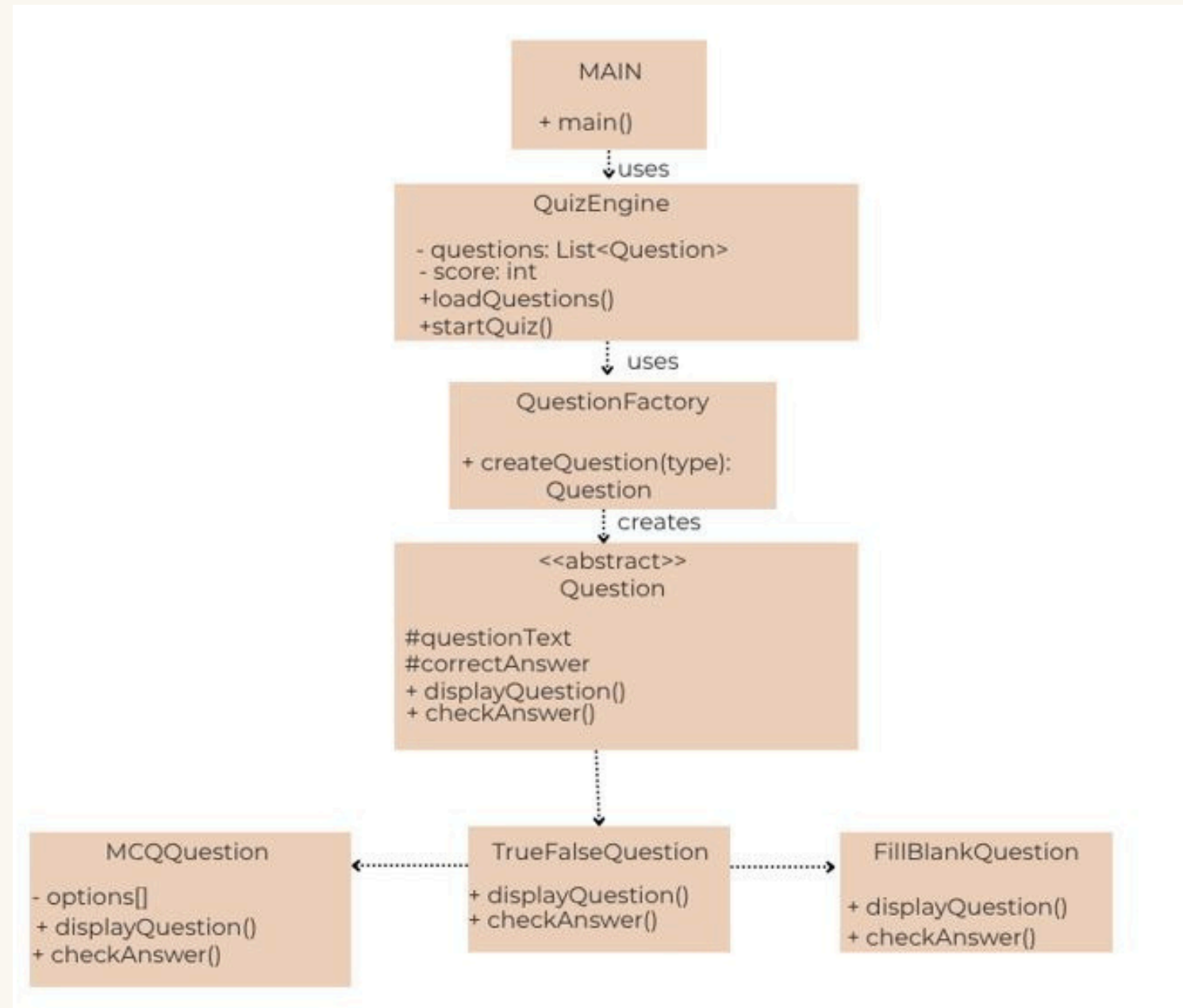
PROJECT TITLE:Online Quiz Engine Using Factory Pattern

**PROBLEM STATEMENT:**Many quiz systems are rigid and difficult to extend when new question types are added. This project builds a flexible quiz engine that supports multiple question formats using Object-Oriented Programming principles.

## Key Classes

- Main – starts the program
- QuizEngine – manages quiz flow and score
- QuestionFactory – creates question objects
- Question – abstract base class
- MCQQuestion / TrueFalseQuestion / FillBlankQuestion – specific question types

## PROJECT STRUCTURE:



## PACKAGE STRUCTURE:

src/

|—— Main.java

|—— QuizEngine.java

|—— Question.java

|—— QuestionFactory.java

|—— MCQQuestion.java

|—— TrueFalseQuestion.java

|—— FillBlankQuestion.java

# OOP Concepts used:

## **Abstraction**

Implemented using the abstract class Question which defines common methods like `displayQuestion()` and `checkAnswer()`.

## **Inheritance**

Classes MCQQuestion, TrueFalseQuestion, and FillBlankQuestion inherit from the Question class.

## **Polymorphism**

Each subclass overrides `displayQuestion()` and `checkAnswer()` so each question type behaves differently.

## **Design Pattern**

Factory Pattern is used in QuestionFactory to create different question objects dynamically.

# Technical Highlight

Design Decision: Using a Factory Pattern for question creation.

Code logic

```
public void loadQuestions() {  
    questions.add(QuestionFactory.createQuestion("MCQ"));  
    questions.add(QuestionFactory.createQuestion("TRUEFALSE"));  
    questions.add(QuestionFactory.createQuestion("FILLBLANK"));  
}
```

## Why it was chosen:

Keeps the system flexible

Makes it easy to add new question type

Avoids modifying existing code

# OUTPUT DEMO

Output for any incorrect answer:

```
Which language is platform independent?  
A. Java  
B. Python  
C. C++  
D. Ruby  
Your Answer: B  
Wrong Answer!
```

Output for any correct answer:

```
Java supports Object Oriented Programming. (True/False)  
Your Answer: True  
Correct!  
  
Java was developed by _____.  
Your Answer: James Gosling  
Correct!  
  
Final Score: 2/3
```

## Error Handling:

The program starts the quiz, displays different types of questions, collects user answers, and calculates the final score. The system also handles incorrect inputs without crashing.

If the answer is correct the program starts the quiz, program prints "Correct!"  
The score increases by 1





